

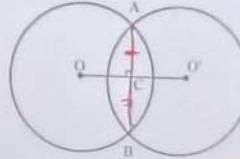


Properties related to Circle

1. The perpendicular from the centre of a circle to a chord bisects the chord.



2. If two circles intersect at two points then the line segment joining their centres is the perpendicular bisector of common chord.

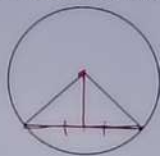


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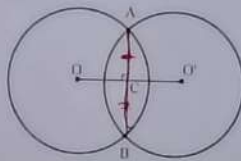


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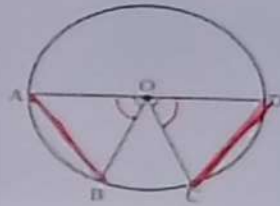


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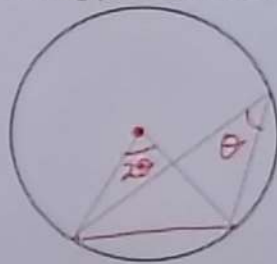
3. Equal chords of a circle subtend equal angle at the centre.



If $AB = CD$ then $\angle AOB = \angle COD$



4. The angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the remaining part of the circle.



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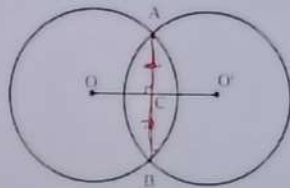


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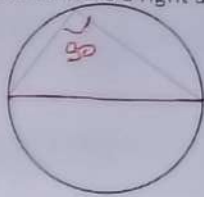
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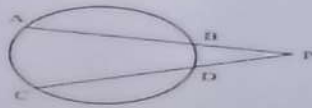
5. The angle in a semi-circle is a right angle.



6. If two chords intersect each other internally or externally then product of their segments are equal.



$$PA \times PB = PC \times PD$$



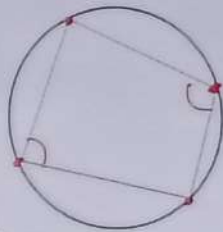
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Cyclic Properties of Circle

A cyclic quadrilateral is called cyclic quadrilateral if its all vertices lie on a circle.



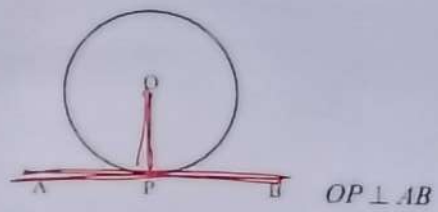
1. The sum of either pair or opposite angles of a cyclic quadrilateral is 180 degree .

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TANGENT AND ITS PROPERTIES

1. A tangent to a circle is perpendicular to the radius through the point of contact.

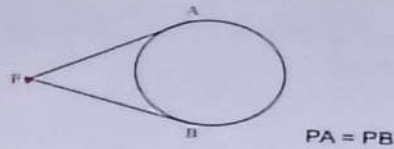


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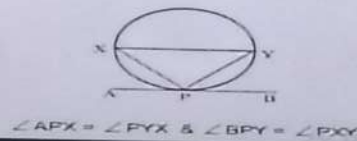
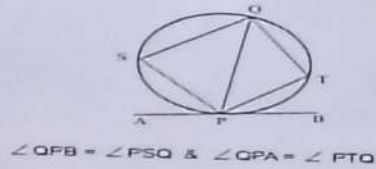




2. From any point outside a circle two tangents can be drawn to it and they are equal in length.



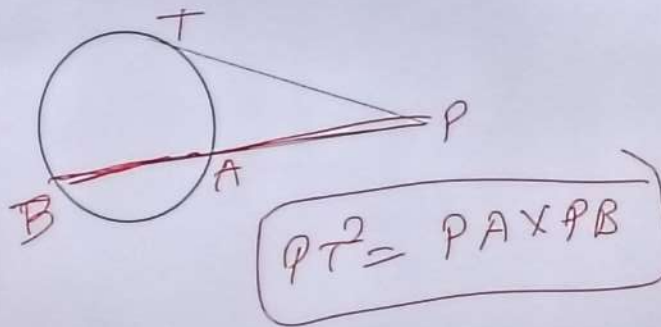
3. If a line touches a circle and from the point of contact a chord is drawn, the angles between the tangent and the chord are respectively equal to the angles in the corresponding alternate segment.



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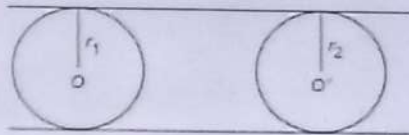
4. If a chord and a tangent intersect externally then the product of the lengths of the segments of the chord is equal to the square of the length of the tangent from the point of contact to the point of intersection.



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The length of a direct common to two circles is:



Length of direct common tangents is

$$= \sqrt{(\text{Distance between their centres})^2 - (r_1 - r_2)^2}$$

where r_1 and r_2 are the radii of the circles

$$= \sqrt{(OO')^2 - (r_1 - r_2)^2}$$

The length of a transverse common tangent to two circles is

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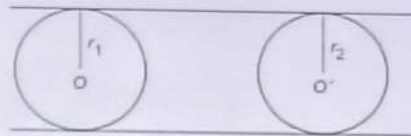
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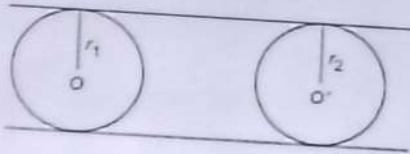
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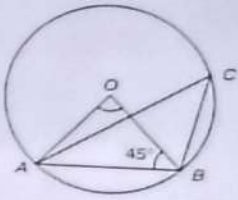
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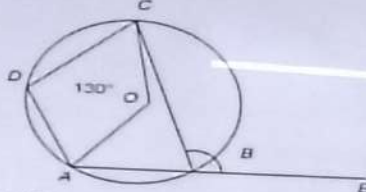


In the given figure, O is the centre of the circle. $\angle ABO = 45^\circ$. Find the value of $\angle ACB$.



- (a) 60°
- (b) 75°
- (c) 90°
- (d) None of these

In the given figure, $\angle AOC = 130^\circ$, where O is the center. Find $\angle CBE$.

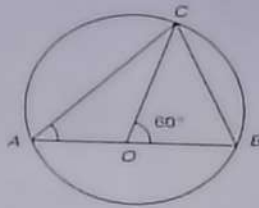


- (a) 100°
- (b) 70°
- (c) 115°
- (d) 130°

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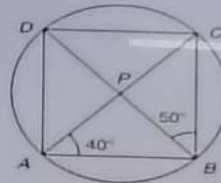


In the given figure, $\angle COB = 60^\circ$, AB is the diameter of the circle. Find $\angle ACO$.



- (a) 20° (b) 30°
(c) 35° (d) 40°

In the given figure, $ABCD$ is a cyclic quadrilateral and the diagonals bisect each other at P . If $\angle CBD = 50^\circ$ and $\angle CAB = 40^\circ$, then $\angle BCD$ is:



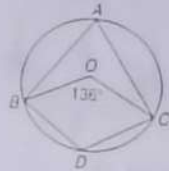
- (a) 60° (b) 75°
(c) 90° (d) 105°



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In the figure, $ABDC$ is a cyclic quadrilateral with O as centre of the circle. If $\angle BOC = 136^\circ$, find $\angle BDC$



- (a) 110° (b) 112°
(c) 109° (d) 115°

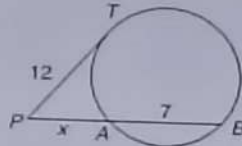


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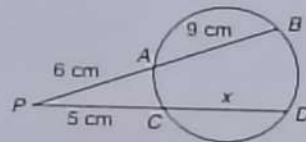


Find the value of x in the given figure.



- (a) 16 cm (b) 9 cm
(c) 12 cm (d) 7 cm

Find the value of x in the given figure.



- (a) 13 cm (b) 12 cm
(c) 16 cm (d) 15 cm



Polygon



Polygons can broadly be divided into two types:

- Regular polygons: Polygons with all the sides and angles equal.
- Irregular polygons: Polygons in which all the sides or angles are not of the same measure.



Things to do

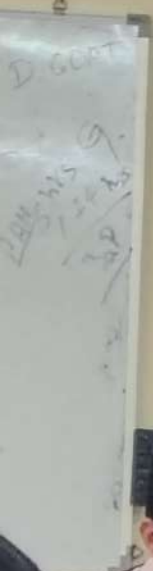
- ☒ Have you seen...
- ☒ Hopes...
- ☒ LearnCo...
- ☒ Indica...
- ☒ Interv...
- ☒ Geostat...
- ☒ Unloaded pr...
- ☒ Unlaid pr...
- ☒ Avoided d... for 4.5 hrs
- ☒ Practiced un...



Properties :-

- ❖ Sum of all the ^{interior} angles of a polygon with n sides = $(n - 2) \times 180^\circ$
- ❖ Sum of all exterior angles = 360°
- ❖ No. of sides = $360^\circ / \text{exterior angle}$. (Note: This property is true only for regular polygons)
- ❖ Area = $(ns^2 / 4) \times \cot(180/n)$; where s = length of side, n = no. of sides.

$$I + E = 180$$



Each interior angle of a regular polygon is 140° . Then the number of sides is:

- (a) 6 (b) 8 (c) 9 (d) 12

Each interior angle of a regular octagon is:

- (a) 90° (b) 115° (c) 125° (d) 135°

The sum of the interior angles of a polygon is 1440° . The number of sides of the polygon is:

- (a) 8 (b) 10 (c) 12 (d) 14